

Mitty Yu

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SUMMARY

A Data Scientist and Automation Specialist combining industrial engineering and data science expertise to create innovative solutions in logistics and operations, expanding horizon into other fields. Proven track record of designing machine learning and deep learning solutions, agentic workflows, optimization, and simulation models at scale. Exceptional at leading cross-functional teams, exercising technical ownership, and navigating highly ambiguous constraints to deliver high-impact engineering solutions.

TECHNICAL SKILLS

Languages: Python (PyTorch, Pandas, NumPy, Scikit-learn), Java, SQL, C++, C, C#, R, Julia

Expertise: Machine Learning, Deep Learning, Statistics, Data Cleaning & Analysis, Causal inference, Optimization

Tools & Cloud: AWS, Google Cloud, Databricks, Tableau, Git, AnyLogic simulation, Bitbucket

CERTIFICATIONS

Project Management Institute

Nov. 2025

Certified Associate in Project Management (CAPM)

EDUCATION

Stanford University

Sep. 2022 – Apr. 2024

MS in Computational and Mathematical Engineering (ICME), Data Science (GPA: 3.7)

Stanford, CA

Relevant Coursework: Software development, CS229 Machine Learning, CS230 Deep Learning, Computer Vision, Natural language processing, Mining massive dataset, Regression & ANOVA, Statistical Inference

National Tsing Hua University

Sep. 2018 – Jun. 2022

BS in Industrial Engineering & Management, minor in Computer Science (GPA: 3.9)

Hsinchu City, Taiwan

PROFESSIONAL EXPERIENCE

Data Scientist

Jul. 2024 – Present

Applied Materials

Santa Clara, CA

Technical lead and developer delivering end-to-end ML, simulation, and optimization products from vague operational challenges. Drove millions in savings by translating ambiguous bottlenecks into deployed system.

- Architected and deployed the organization's first suite of eight Java-based predictive digital twins to AnyLogic cloud; scaled these high-fidelity models to drive global adoption of simulation and optimize high-concurrency logistics workflows, generating **\$2M+** in projected savings with **95%** accuracy in throughputs and risk reduction.
- Built an end-to-end workforce planning framework combining XGBoost, regression, and time-series models with discrete-event simulation, optimizing cross-functional warehouse staffing allocations.
- Developed an end-to-end automated 3D-slotting optimizer from framing to deployment using Claude Code and Databricks; engineered a greedy heuristic synchronizing SKU velocity with spatial constraints, increasing throughput by 23% and generating \$200k in quarterly savings.
- Developed **end-to-end ETL pipelines** with Python and SQL on Databricks to analyze inventory aging anomalies, successfully identified **\$50k+** in obsolete stock and migrated insights to real-time dashboards.
- Engineered a **computer vision** algorithm utilizing Python and YOLOv8 to automate inventory counting via mobile camera, with a **99% counting accuracy** for common parts.
- Partnered with **cross-functional experts** to identify operational pain-points; translated complex data into actionable recommendations that optimized product delivery, automation, and reduced operational risk by 80%.
- Led a technical team through the **full project lifecycle** using **CAPM** methodologies; used Git, Bitbucket, and Smartsheet to streamline development workflows, driving a **15%** increase in efficiency.

Software Algorithm Engineer Intern

Jun. 2023 – Sep. 2023

ASML

San Jose, CA

- Developed a defect classification model achieving **99.62% accuracy** to automate anomaly detection within semiconductor production metrology workflows; implemented Saliency Maps to interpret CNN feature attribution and drive model selection.

- Conceptualized a patent-eligible method for loading image patches, showcasing strong problem-solving skills and innovation. Developed custom DataLoaders for feature extraction from single-beam image patches with CNNs.
- Built an end-to-end t-SNE evaluation pipeline integrated with a KNN classifier to analyze high-dimensional feature spaces; benchmarked dimensionality reduction tools like PCA, t-SNE, and PaCMAP to optimize data visualization.

ACADEMIC PROJECTS

Representation Learning for Land Cover Classification

Jun. 2023

Computer vision, Image processing, Python, PyTorch, Pandas, Scikit-learn, AWS, Git

[Report link](#)

- Engineered a self-supervised representation learning framework with "Tile2Vec" architecture to improve land cover classification accuracy and efficiency on large-scale satellite imagery with limited labeled data, supporting UN Sustainable Development Goal 15 (Life on Land).
- Evaluated and benchmarked diverse backbone architectures, including ResNet, GoogLeNet, and Vision Transformers; optimized training workflows to deploy a custom GoogLeNet-V1 backbone that reduced parameters by 50% and accelerated training velocity by 7.5x while maintaining comparable embedding quality.

Supplementing The Ocean Cleanup's Global Dataset with Named Entity Recognition

Mar. 2023

Natural Language Processing, Python, PyTorch, Pandas, AWS, Git, Text analysis, Web Scraping

[Report link](#)

- Awarded **2nd** place for Best Poster at the **Stanford ICME XPO Research Symposium**.
- Designed and deployed an end-to-end Named Entity Recognition pipeline for The Ocean Cleanup to extract beach cleanup activities metrics from unstructured social media and web text, improving the tracking of global coastal plastic transport and sinks.
- Experimented and tuned Transformer, BERT, and LSTM architectures using PyTorch to achieve a 94% F1-score for automated global data extraction.

Ingredients and Recipes recommendation system with Machine Learning

Dec. 2023

Machine learning, Deep learning, Python, Pytorch, Naïve Bayes, Word2Vec, Google Cloud

[Report link](#)

- Created an intuitive culinary recommendation system by converting unstructured recipe metadata into high-performance semantic search representations to enhance the kitchen experience.
- Developed an end-to-end data pipeline processing 10k+ recipe records on Google Cloud, implementing custom category mapping, Word2Vec item embedding, and applied Principal Component Analysis (PCA) to compress high-dimensional sparse representations while maintaining analytical model invertibility. Mitigated extreme rating skewness by introducing inverse-frequency sample weights to stabilize loss optimization.

Precipitation Nowcasting From Radar Images Using Deep Learning

Jun. 2023

Computer vision, Python, Pytorch, Image processing, Data visualization, AWS, Git, GitHub

[Report link](#)

- Enhanced localized severe weather predictability by engineering deep learning encoding-forecasting models capable of capturing highly non-linear, spatial-temporal storm movements.
- Built a heterogeneous deep learning pipeline using a 3-layer encoder-forecaster framework; integrated Trajectory GRU (TrajGRU) and Convolutional LSTM (ConvLSTM) layers to balance sequence-dependent temporal modeling with location-variant spatial tracking.
- Evaluated generative structural accuracy using Gradient Difference Loss (GDL) alongside pixel-wise regression metrics (MSE/MAE); proved that structural-loss constraints significantly reduce spatial blurring over a rolling validation window.

PERSONAL PROJECTS

AI-powered Life OS

Jun. 2026

Google Antigravity, Python, Google Gemini API, python-telegram-bot, Obsidian Vault Engine

- Built a personal AI assistant on Telegram that eliminates manual logging for ideas, personal projects tracking, meals, workouts, journaling, and to-do lists by automatically updating an Obsidian notebook vault in real time.
- Added support for video parsing, automatically capture ideas from shared Facebook/Instagram reels with AI.